National University of Computer & Emerging Sciences

Homework # 1(a)

Frequency Distribution

1. Consider the data below, that represents the number of hours spent on social media by 30 individuals in a single day:

$$\{0, 1, 2, 3, 1, 4, 5, 2, 3, 4, 6, 3, 2, 5, 6, 7, 1, 3, 2, 4, 6, 5, 3, 7, 6, 2, 1, 4, 3, 5\}$$

   - Clearly describe the variable (hours spent on social media) and its importance in the given context.
   - Identify whether the data is *qualitative* or *quantitative*.
   - If quantitative, specify whether it is *discrete* or *continuous*.

2. **Use Sturges' Rule:** Calculate the number of bins (k) for creating the frequency distribution table using Sturges' Rule:

$$k = 1 + 3.322 \log_{10}(n)$$

   Here, $n = 30$ is the number of observations in the dataset. Use the calculated value of k to determine the class intervals for the frequency distribution.

3. **Frequency Distribution Table:**
   - Create a frequency distribution table based on the calculated number of bins.
   - Include columns for class intervals, class boundaries, frequencies, midpoints, and cumulative frequencies.

Python Code Example

Below is an example of how you can use Python with Matplotlib to create a histogram. Use this as a guide for creating your graphs.

```
import matplotlib.pyplot as plt

# Sample data
data = [1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 6, 6, 6, 6, 7, 7, 8]

# Create the histogram
plt.hist(data, bins=7, color='blue', edgecolor='black')
plt.title('Histogram of Sample Data')
plt.xlabel('Data Values')
plt.ylabel('Frequency')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```